SIMPLE OVERVIEW OF CONFORMAL PREDICTION

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WHY PREDICTION INTERVALS?

- ML models return point predictions $\hat{f}(x)$ but quantify no uncertainty
- Bayesian posteriors, bootstrap, quantile regression: all need model assumptions or asymptotics
- Conformal prediction (CP) returns a set $\mathcal{C}(x)$ with a hard guarantee

$$\underbrace{\mathbb{P}\Big(Y_{n+1} \in \mathcal{C}(X_{n+1})\Big)}_{\text{probability the true label is in the set}} \geq \underbrace{1-\alpha}_{\text{user-chosen miscoverage level}}$$

- Distribution-free (**no assumption on the data law**), finite-sample (**works at any** n), model-agnostic (**wraps any predictor**)
- Only assumption: exchangeability of $(X_i, Y_i)_{i=1}^{n+1}$ – the data are interchangeable

BACKGROUND: THREE TERMS THAT DO ALL THE WORK

- Predictor $\hat{f} : \mathcal{X} \to \mathcal{Y}$ — any black-box model (linear, XGBoost, deep network). CP never inspects $\hat{f}$'s internals.
- Exchangeability — the joint distribution of $(Z_1, \ldots, Z_{n+1})$ is invariant under permutation. **Strictly weaker than i.i.d.**: any reshuffling of i.i.d. data is exchangeable.
- Calibration set $\{(X_i, Y_i)\}_{i=1}^{n}$ — a held-out subset, disjoint from training data. Used to compute the cutoff.
- Miscoverage level $\alpha \in (0, 1)$ — the maximum allowed failure probability. $\alpha = 0.1 \Rightarrow 90\%$ prediction sets.
- Nonconformity score $s(x, y) \in \mathbb{R}$ — "how surprising is the pair (x, y) given $\hat{f}$?" Larger means more surprising. The user designs this.

THE RECIPE – STEP 1: HEURISTIC UNCERTAINTY

- Start with any predictor $\hat{f}$. Nothing else is required of it.
- Examples:
  - Regression: $\hat{f}(x)$ is a point estimate $\in \mathbb{R}$
  - Classification: $\hat{f}_y(x) \in [0,1]$ is the softmax probability of class y
  - Quantile regression: $\hat{f}$ returns the conditional $\alpha/2$ and $1-\alpha/2$ quantiles
- No assumption on $\hat{f}$. It can be miscalibrated — its raw probabilities can be meaningless — and CP still works.
- The role of $\hat{f}$ is just to provide a heuristic sense of where the true y might lie. CP then calibrates that heuristic into a valid set.

THE RECIPE – STEP 2: NONCONFORMITY SCORE $s(x, y)$

- Design a function $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ measuring "how surprising is (x, y) given $\hat{f}$?"
- Regression — absolute residual:

$$\underbrace{s(x, y)}_{\text{score}} = \underbrace{|\, y - \hat{f}(x) \,|}_{\substack{\text{distance from} \\ \text{point prediction}}}$$

- Classification — one minus the true-class probability:

$$s(x, y) = 1 - \hat{f}_y(x)$$

- Score design = user's choice. The coverage guarantee survives *any* choice. The choice affects set shape and size, not validity.

THE RECIPE – STEP 3: CALIBRATION QUANTILE $\hat{q}$

Compute calibration scores $s_i = s(X_i, Y_i)$ for $i = 1, \ldots, n$ and take the *adjusted* empirical quantile:

$$\underbrace{\hat{q}}_{\substack{\text{cutoff that}\\ \text{calibrates the set}}} = \mathrm{Quantile}\Big(\underbrace{\{s_i\}_{i=1}^{n}}_{\substack{\text{scores on}\\ \text{held-out data}}} \; ; \; \underbrace{\frac{\lceil (n+1)(1-\alpha) \rceil}{n}}_{\substack{\text{slightly above } 1-\alpha;\\ \text{the } +1 \text{ corrects for the}\\ \text{unseen test point}}} \Big)$$

- The $(n+1)$ correction is why CP works at *finite n*. It accounts for the test point becoming the $(n+1)$-th observation.
- Without it (just $\lceil n(1-\alpha) \rceil$): you under-cover at small n.
- With it: coverage is at least $1-\alpha$ exactly, not asymptotically.

THE RECIPE – STEP 4: PREDICTION SET $\mathcal{C}(X_{n+1})$

$$\underbrace{\mathcal{C}(X_{n+1})}_{\text{set we hand to the user}} = \left\{ y \in \mathcal{Y} : \underbrace{s(X_{n+1}, y)}_{\text{score of candidate } y \text{ against } \hat{f}} \le \underbrace{\hat{q}}_{\text{calibrated cutoff}} \right\}$$

- All y values that the score function declares not-too-surprising.
- Regression with absolute-residual score: $\mathcal{C}(x) = [\hat{f}(x) - \hat{q},\ \hat{f}(x) + \hat{q}]$ — a fixed-width band.
- Classification with one-minus-softmax score: $\mathcal{C}(x) = \{y : 1 - \hat{f}_y(x) \le \hat{q}\}$ — a set of labels.
- In both cases the set is automatic once you've fixed s and computed $\hat{q}$.

THE COVERAGE THEOREM

Theorem (Marginal coverage)

If $(X_i, Y_i)_{i=1}^{n+1}$ *are exchangeable, then*

$$\underbrace{\mathbb{P}\big(Y_{n+1} \in \mathcal{C}(X_{n+1})\big)}_{\text{probability the true label is in the set}} \in \Big[\ \underbrace{1-\alpha}_{\text{lower bound; always achieved}}\ ,\ \underbrace{1-\alpha+\tfrac{1}{n+1}}_{\text{upper bound; discreteness slack}}\ \Big].$$

- Always at least $1-\alpha$ — the lower bound holds for any data law and any $\hat{f}$.
- No more than $1/(n+1)$ over – at $n = 100$ the slack is $\sim 1\%$; at $n = 1000$ it's $\sim 0.1\%$.
- Marginal: averaged over the joint draw of test point *and* calibration set. Conditional coverage is a separate (and harder) question.

PROOF SKETCH + BETA COROLLARY

- Three-sentence proof: (1) Exchangeability means the test point is interchangeable with any calibration point. (2) So the rank of $s(X_{n+1}, Y_{n+1})$ among all $n+1$ scores is uniform on $\{1, \ldots, n+1\}$. (3) Setting the cutoff at rank $\lceil (n+1)(1-\alpha) \rceil$ bounds the probability that the test rank exceeds it.
- Corollary – Beta-distributed coverage: for a *fixed* calibration set, the realised coverage is itself random:

$$\underbrace{C}_{\substack{\text{realised coverage} \\ \text{for this cal. set}}} \sim \mathrm{Beta}\Big(\underbrace{\lceil (n+1)(1-\alpha) \rceil}_{\text{shape 1}}, \underbrace{n+1-\lceil (n+1)(1-\alpha) \rceil}_{\text{shape 2}} \Big)$$

- Rule of thumb: $n \approx 1{,}000$ keeps realised coverage within $\pm 1\%$ of nominal with high probability. At $n = 100$ the standard deviation is $\sim 3\%$ — meaningfully wide.

SCORE CATALOGUE – CLASSIFICATION

- Plain softmax score: $s(x, y) = 1 - \hat{f}_y(x)$. Fixed-size sets (every class either in or out at the cutoff).
- APS (Adaptive Prediction Sets, Sadinle–Lei–Wasserman 2019): rank softmax probabilities, accumulate until total mass $\geq \hat{q}$:

$$s(x, y) = \sum_{k : \hat{f}_k(x) \geq \hat{f}_y(x)} \hat{f}_k(x)$$

  Sets grow on hard inputs and shrink on easy inputs — input-difficulty adaptive.

- RAPS (Regularised APS, Angelopoulos–Bates 2022): adds a penalty term to prevent pathologically large sets on tail classes.
- All three: same marginal coverage theorem; differ only in set shape.

SCORE CATALOGUE – REGRESSION AND CQR

- Plain absolute-residual: $s(x, y) = |y - \hat{f}(x)|$. Constant-width intervals.
- σ-scaled: $s(x, y) = |y - \hat{f}(x)|/\hat{\sigma}(x)$ with a separately fitted scale $\hat{\sigma}$. Locally adaptive width.
- CQR (Conformalised Quantile Regression, Romano–Patterson–Candès 2019): fit a quantile regressor returning $\hat{q}_{\alpha/2}(x)$ and $\hat{q}_{1-\alpha/2}(x)$, define

$$s(x, y) = \max\Big(\underbrace{\hat{q}_{\alpha/2}(x) - y}_{\substack{\text{undershoot} \\ \text{of lower quantile}}} , \underbrace{y - \hat{q}_{1-\alpha/2}(x)}_{\substack{\text{overshoot} \\ \text{of upper quantile}}} \Big)$$

  Intervals are asymmetric and adapt to heteroscedasticity (locally varying width).
- All three: same coverage guarantee; CQR usually narrowest on heteroscedastic data.

CONFORMAL RISK CONTROL (CRC)

- Replace the indicator coverage loss with any bounded monotone loss $L(\lambda) \in [0,1]$. Tune the cutoff λ so that the expected loss meets a budget:

$$\underbrace{\mathbb{E}[L(\lambda)]}_{\substack{\text{expected loss}\\ \text{at cutoff } \lambda}} \leq \underbrace{\alpha}_{\substack{\text{user-chosen}\\ \text{risk budget}}}$$

- Applications:
  - FNR control in image segmentation (limit missed pixels)
  - FDR control in multi-label classification
  - Mass coverage in distributional prediction
- Learn-Then-Test (LTT): extends CRC to non-monotone risks via multiple-testing corrections on a grid of λ.
- Generalises the coverage recipe — any time you can name a controllable bounded risk,

BEYOND SPLIT CP – COMPUTE/STATISTICS SPECTRUM

- Split CP: one fit; calibration set is wasted for training. **Default for big data.**
- Full Conformal (Vovk 2005): retrain $\hat{f}$ for *each candidate* $y \in \mathcal{Y}$ on the augmented dataset. Exact, but $O(|\mathcal{Y}|)$ refits per test point — usually infeasible.
- Cross-Conformal / CV+: K-fold variant. Modest extra compute, recovers most of the lost data efficiency.
- Jackknife+ (Barber–Candès–Ramdas–Tibshirani 2021):

$$\underbrace{\mathbb{P}\left(Y_{n+1} \in \mathcal{C}^{\mathrm{J+}}(X_{n+1})\right)}_{\substack{\text{coverage of} \\ \text{Jackknife+ interval}}} \geq \underbrace{1 - 2\alpha}_{\substack{\text{factor-2 looser bound;} \\ \text{LOO from one fit via} \\ \text{comparison-matrix argument}}}$$

  Distribution-free, no retraining at test time. **Default when data is scarce.**
- Rule of thumb: lots of data $\Rightarrow$ Split. Scarce data $\Rightarrow$ Jackknife+.

TIME SERIES: WHERE EXCHANGEABILITY BREAKS

Two structurally distinct failure modes:

1. Temporal dependence — the process has memory:
   - AR / ARMA: Y_t depends on past values $Y_{t-1}, Y_{t-2}, \ldots$
   - GARCH: conditional variance $\mathrm{Var}(Y_t \mid Y_{<t})$ depends on past
   - Latent state, regime switching, hidden Markov…

   Even stationary series violate exchangeability if they have dependence.

2. Distribution shift — the joint law itself moves:
   - Abrupt breaks (regulatory changes, regime shifts, 2020/2021 crises)
   - Gradual drift (slow trends, evolving user behaviour)
   - Seasonal cycles (daily, weekly, annual)

Either failure breaks the finite-sample coverage guarantee. Different fixes target different failures.

THE FOUR-FAMILY TAXONOMY (STOCKER ET AL. 2025)

Every modern CP-for-time-series method modifies exactly one object of the SCP procedure:

1. Reweight calibration data (→ **WCP**, **NexCP**): down-weight stale scores via a recency kernel
2. Refresh residual pool (→ **EnbPI**, **SPCI**): bootstrap LOO residuals and slide a fresh window
3. Adapt α_t online (→ **ACI**, **AgACI**, **Conformal PID**): update the target miscoverage after each observation
4. Randomise over blocks (→ **Block CP**): shuffle contiguous chunks to restore block-level exchangeability

Under abrupt mean shift: SCP, Block-SCP, WCP-window fail to cover; ACI, EnbPI, WCP-exp, WCP-linear self-correct.

Hybrid methods compose families (e.g., ACI + SPCI conditional quantiles).

ENBPI (XU & XIE 2023) – BOOTSTRAP LOO

- Train B bootstrap predictors $\{\hat{f}^{(b)}\}_{b=1}^{B}$. For each training index t, aggregate only the bootstrap models that didn't see t:

$$\underbrace{\hat{f}^{-t}(x)}_{\text{LOO estimator at index } t} = \frac{1}{|\{b : t \notin \mathcal{I}_b\}|} \sum_{b:t\notin\mathcal{I}_b} \underbrace{\hat{f}^{(b)}(x)}_{\text{bootstrap model}} .$$

- LOO residuals: $\hat{\varepsilon}_t = Y_t - \hat{f}^{-t}(X_t)$.
- Test-time interval: sliding-window quantile of recent residuals; asymmetric via inner optimisation over $\beta \in [0, \alpha]$.
- No data split, no retraining. One ensemble fit upfront; $O(1)$ per test point.
- Asymptotic conditional + marginal coverage under β-mixing — rare for CP-on-time-series.

ACI (GIBBS–CANDÈS 2021) – ONLINE α-UPDATE

At each time t, observe whether the interval covered, then update the target:

$$\underbrace{\alpha_{t+1}}_{\substack{\text{next target}\\\text{miscoverage}}} = \underbrace{\alpha_t}_{\substack{\text{current}\\\text{target}}} + \underbrace{\gamma}_{\substack{\text{learning}\\\text{rate}}} \Big(\underbrace{\alpha}_{\substack{\text{nominal}\\\text{target}}} - \underbrace{\mathbb{1}\{Y_t \notin \hat{C}_t\}}_{\substack{\text{1 if missed,}\\\text{0 if covered}}} \Big)$$

- Cover → tighten ($\alpha_{t+1} > \alpha_t$); miss → loosen ($\alpha_{t+1} < \alpha_t$).
- Long-run miscoverage → α for *any* sequence — adversarial, drifting, exchangeable, all of it.
- γ trade-off: large γ adapts fast but wastes interval width on stationary data; small γ is stable but slow to react.
- AgACI: aggregate ACI instances at multiple γ values via online experts — parameter-free.

COVERAGE HIERARCHY – THREE FLAVOURS, WEAKEST TO STRONGEST

1. Marginal (what standard CP gives):

$$\mathbb{P}\big(Y \in \mathcal{C}(X)\big) \;\geq\; 1 - \alpha \qquad \underbrace{\text{(averaged over } X, Y\text{)}}_{\text{joint test draw}}$$

2. Calibration-conditional (PAC-style, Vovk 2012; Bian–Barber 2023): fix a calibration set; coverage holds w.p. $\geq 1 - \delta$ over calibration draws.
3. X-conditional (what you want):

$$\mathbb{P}\big(Y \in \mathcal{C}(X) \mid X = x\big) \;\geq\; 1 - \alpha \qquad \underbrace{\forall\, x}_{\text{every input}}$$

Approximate substitutes for X-conditional (since the real thing is impossible):

- Class-conditional / group-conditional / Mondrian CP

VOVK IMPOSSIBILITY – THE NEGATIVE RESULT THAT FRAMES THE FIELD

Lemma (Vovk 2012, paraphrased)

Distribution-free X-conditional coverage at non-trivial level is achievable only by prediction sets of infinite Lebesgue measure on a positive-probability subset of $\mathcal{X}$.

- Translation: you cannot have *per-individual* coverage guarantees distribution-free. Full stop.
- Every "approximate conditional" CP method is an honest engineering compromise around this hard fact:
  - **Local adaptivity** (σ-scaled, CQR): intervals that *look right* at each x, even if not formally guaranteed at x.
  - **Partition** (Mondrian / class-cond. / group-balanced): per-cell coverage, not per-point.
  - **PAC slack** (calibration-conditional): coverage that holds w.p. $1 - \delta$ over cal. draws.
  - **Asymptotic** (consistent quantile regression): conditional coverage in the $n \to \infty$ limit, at the cost of distribution-freeness.

TAKE-HOME AND READING ORDER

- CP is a wrapper, not a model — drops on top of any predictor.
- Pick the variant by data regime:
  - **IID, lots of data** ⇒ Split CP
  - **IID, scarce data** ⇒ Jackknife+ or CV+
  - **Time series, stationary** ⇒ EnbPI / SPCI
  - **Time series, drift** ⇒ ACI / AgACI / Conformal PID
  - **Asymmetric / heteroscedastic regression** ⇒ CQR
  - **Missing covariates** ⇒ CP-MDA-Nested
- Libraries: `MAPIE`, `crepes`, `conformal-prediction` (Angelopoulos)
- Reading order:
  1. Angelopoulos & Bates (2022) – *A Gentle Introduction to CP*
  2. Tibshirani (2023) – CMU lecture notes (statistician's view)
  3. Stocker et al. (2025) – *Gentle Introduction to Conformal Time Series Forecasting*